Endogenous Education Choice in Segmented Search Markets

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1 Global Environment

Time is continuous. All agents discount at rate $r > 0$.

Aggregate productivity. Aggregate productivity z_t follows a finite-state continuous-time Markov chain on $\mathcal{Z} = \{z_1, \ldots, z_K\}$ with generator matrix $\mathbf{Q} = (q_{zz'})$. For any value function $V(z)$,

$$(r - \mathbf{Q})V(z) \equiv rV(z) - \sum_{z' \neq z} q_{zz'}\big(V(z') - V(z)\big).$$

2 Workers, Firms, and Types

2.1 Workers

Each worker has an innate type $x \in \mathcal{X} = [0, 1]$, fixed over time.

Worker types are distributed according to CDF L with density ℓ. In applications we assume $x \sim \text{Beta}(a_x, b_x)$.

Market access. Workers can access either:

- the *unskilled* labor market U,
- the *skilled* labor market S.

Training changes *market access only*; it does not change x.

Type cutoff and population partition. For each aggregate state z, a cutoff $\bar{x}(z) \in [0, 1]$ partitions the worker population. Types $x \leq \bar{x}(z)$ participate in the unskilled labor market and are subject to unskilled idiosyncratic shocks and job destruction. Types $x > \bar{x}(z)$ are interpreted as selecting the skilled track whenever they are in unskilled unemployment (i.e. they initiate training immediately upon becoming eligible), and therefore do not contribute to the steady-state stock of unskilled unemployment or unskilled matches. Accordingly, objects in the unskilled block are computed on the truncated type distribution $L(\cdot \mid x \leq \bar{x}(z))$, and objects on the skilled side are computer on the opposite truncation.

2.2 Firms

There are two labor markets.

Unskilled firms. Unskilled firms are homogeneous.

Skilled firms. Each skilled firm has a permanent type $y \in \mathcal{Y} = [0, 1]$, distributed according to CDF Γ with density γ.

3 Training Technology

Eligibility. Only *unskilled unemployed* workers may initiate training.

Cost. Initiating training requires paying a one-time cost $c(x)$, where

$$c'(x) < 0.$$

Completion. Training completes at Poisson rate $\phi > 0$. Upon completion, the worker becomes *skilled unemployed.*

Flow payoffs. Let $b_U(z)$, $b_T(z)$, and $b_S(z)$ denote flow payoffs while unskilled unemployed, training, and skilled unemployed, respectively.

Specification. The cost function is specified as $c(x) = a_c e^{-x}$

4 Production Technologies

Aggregate productivity affects the two labor markets asymmetrically. Let $z \in \mathcal{Z}$ denote the aggregate state. Associated with each state are sector-specific productivity shifters $P_U(z) > 0$ and $P_S(z) > 0$.

Unskilled production. An unskilled match between a worker of type x and a homogeneous firm produces

$$\pi_U(x, z) = P_U(z)\, x.$$

Skilled production. A skilled match between worker type x and firm type y produces

$$\pi_S(x, y, z) = P_S(z)\, f(x, y),$$

where

$$f(x, y) = [\omega x^\rho + (1 - \omega) y^\rho]^{1/\rho}, \qquad \omega \in (0, 1),\ \rho \leq 1,\ \rho \neq 0.$$

5 Unskilled Labor Market

5.1 Matching

Let $\mathcal{U}_U(z)$ and $\mathcal{V}_U(z)$ denote the masses of unskilled unemployed workers and unskilled vacancies in aggregate state z. Market tightness is

$$\theta_U(z) = \frac{\mathcal{V}_U(z)}{\mathcal{U}_U(z)}.$$

Matching follows a Cobb–Douglas technology:

$$M_U(\mathcal{U}_U(z), \mathcal{V}_U(z)) = \mu_U\, \mathcal{U}_U(z)^{\eta_U} \mathcal{V}_U(z)^{1-\eta_U}, \qquad \eta_U \in (0, 1).$$

The vacancy filling rate is

$$q_U(\theta_U(z)) = \mu_U\, \theta_U(z)^{-\eta_U},$$

and the job-finding rate for workers is $\theta_U(z) q_U(\theta_U(z))$.

5.2 Match productivity and shocks

An unskilled match is characterized by productivity $x \in [0, \bar{x}(z)]$ and produces

$$\pi_U(x, z) = P_U(z)\, x.$$

Idiosyncratic productivity shocks arrive at Poisson rate $\lambda_U > 0$. Upon a shock, productivity is redrawn from L truncated to $[0, \bar{x}(z)]$. New matches are created at productivity $x_0 = \bar{x}(z)$.

5.3 Value functions

Let:

- $U_U(x, z)$ denote the value of unskilled unemployment for a worker of type x,
- $T(x, z)$ the value of training,
- $E_U(x, z)$ the value of employment at productivity x,
- $J_U(x, z)$ the value of a filled job with productivity x.
- $V_U(x)$ the value of a vacancy at productivity x.

Unemployment value. An unemployed worker may either search in the unskilled market or initiate training. While training, the worker pays a cost that is decreasing in type x i.e. it is easier for more productive workers to get training. Training has no impact on productivity, and only serves as a way for workers to access the skilled labour market. Training is completed stochastically at rate ϕ, when that happens, the worker joins the skilled market as unemployed. While searching, the worker gets unemployment benefits $b_U(z)$, and meets new jobs at rate $\theta_U(z) q_U(\theta_U(z))$.

$$U_U(x, z) = \max\left\{U_U^{\text{search}}(z), U_U^{\text{train}}(x, z)\right\}.$$

Where:

$$U_U^{\text{train}}(x, z) = -c(x) + T(x, z)$$

While training,

$$(r - \mathbf{Q})T(x, z) = b_T(z) + \phi\big(U_S(x, z) - T(x, z)\big).$$

While searching,

$$(r - \mathbf{Q})U_U^{\text{search}}(z) = b_U(z) + \theta_U(z) q_U(\theta_U(z))\big(E_U(\bar{x}(z), z) - U_U^{\text{search}}(z)\big).$$

Existence of the market-access cutoff. Fix an aggregate state z. Define the value difference between initiating training and remaining unskilled unemployed as

$$\Delta_T(x, z) := U_U^{\text{train}}(x, z) - U_U^{\text{search}}(z) = -c(x) + T(x, z) - U_U^{\text{search}}(z).$$

Assumption 1 (Monotone training incentives)**.** *For each $z \in \mathcal{Z}$, the function $x \mapsto \Delta_T(x, z)$ is continuous and strictly increasing on* $[0, 1]$.

Assumption 1 is satisfied if training costs $c(x)$ are strictly decreasing and the continuation value $T(x, z)$ is weakly increasing in x.

Cutoff characterization. Suppose there exist types $x^-(z), x^+(z) \in [0,1]$ such that

$$\Delta_T\big(x^-(z), z\big) \le 0, \qquad \Delta_T\big(x^+(z), z\big) \ge 0.$$

Then there exists a unique cutoff $\bar{x}(z) \in [0,1]$ satisfying

$$U_U^{\text{search}}(z) = -c(\bar{x}(z)) + T(\bar{x}(z), z),$$

with

$$\Delta_T(x,z) \begin{cases} < 0, & x < \bar{x}(z), \\ = 0, & x = \bar{x}(z), \\ > 0, & x > \bar{x}(z). \end{cases}$$

Thus, workers with $x > \bar{x}(z)$ strictly prefer to initiate training upon becoming unskilled unemployed, while workers with $x \le \bar{x}(z)$ remain in the unskilled labor market.

Employment cutoff. Fix an aggregate state z. Define the *employment–unemployment value gap*

$$\Delta_U(x,z) \;:=\; E_U(x,z) - U_U(x,z), \qquad x \in [0, \bar{x}(z)].$$

We say that a worker of type x accepts (or retains) an unskilled job in state z whenever $\Delta_U(x,z) \ge 0$.

Assumption 2 (Single-crossing in x)**.** *For each $z \in \mathcal{Z}$, the map $x \mapsto \Delta_U(x,z)$ is continuous on $[0, \bar{x}(z)]$ and (weakly) increasing.*

Under Assumption 2, if there exist types $x^-(z), x^+(z) \in [0, \bar{x}(z)]$ such that

$$\Delta_U\big(x^-(z), z\big) \le 0 \qquad \text{and} \qquad \Delta_U\big(x^+(z), z\big) \ge 0,$$

then there exists a reservation cutoff $x_r(z) \in [0, \bar{x}(z)]$ satisfying

$$\Delta_U\big(x_r(z), z\big) = 0, \qquad \text{i.e.} \qquad E_U\big(x_r(z), z\big) = U_U\big(x_r(z), z\big),$$

and the acceptance set is an interval:

$$\Delta_U(x,z) \ge 0 \iff x \in [x_r(z), \bar{x}(z)], \qquad \Delta_U(x,z) < 0 \iff x \in [0, x_r(z)).$$

If, in addition, $x \mapsto \Delta_U(x,z)$ is strictly increasing, then $x_r(z)$ is unique.
Proof sketch. Continuity implies existence of a root of $\Delta_U(\cdot, z)$ by the intermediate value theorem. Monotonicity (single-crossing) implies the set $\{x : \Delta_U(x,z) \ge 0\}$ is an upper set, hence of the form $[x_r(z), \bar{x}(z)]$. □

Employment. A worker gets wage $w_U(x,z)$, and gets idiosyncratic productivity shocks at rate λ_U, thus x is redrawn from the unskilled labour pool. Let x' be the post-shock productivity. If $x' < x_r$, the worker loses the job and moves to unemployment. Otherwise, if $x' \ge x_r$, the worker gets employed at new wage corresponding to the new productivity level.

$$\big((r - \mathbf{Q}) + \lambda_U\big) E_U(x,z) = w_U(x,z) + \lambda_U \Bigg[\int_0^{x_r(z)} U_U(x', z)\, \mathrm{d}\widetilde{L}_z(x') + \int_{x_r(z)}^{\bar{x}(z)} E_U(x', z)\, \mathrm{d}\widetilde{L}_z(x') \Bigg],$$

where $\widetilde{L}_z$ denotes the distribution L truncated to $[0, \bar{x}(z)]$.

Firm value. An employed worker's production is proportional to their type and the state of the economy, be it $\pi_U(x,z) = P_U(z)x$, gets paid wage $w_U(x,z)$, and gets idiosyncratic productivity shocks at rate λ_U.

$$\big((r-\mathbf{Q})+\lambda_U\big)J_U(x,z) = P_U(z)x - w_U(x,z) + \lambda_U \int_{x_r(z)}^{\bar{x}(z)} J_U(x',z)\,\mathrm{d}\widetilde{L}_z(x').$$

Vacancy value. Firms post vacancies at a cost k_U. These vacancies meet new jobs at the vacancy filling rate $q_U(\theta_U(z))$. When a vacancy is filled, it is taken out of the vacancy pool, and a job is created at productivity $\bar{x}(z)$.

$$(r-\mathbf{Q})V_U(z) = -k_U + q_U(\theta_U(z))\big[J_U(\bar{x}(z),z) - V_U\big]$$

Firms keep posting vacancies until $V_U(z) = 0$ at equilibrium (free entry condition).

5.4 Wages and job destruction

Wages. Wages are determined by Nash bargaining with worker bargaining weight $\beta_U \in (0,1)$:

$$\beta_U J_U(x,z) = (1-\beta_U)\big(E_U(x,z) - U_U(x,z)\big).$$

Endogenous job destruction. $J_U(x,z)$ is increasing in x. Thus, $\exists\, x_r(z)$ such that:

$$J_U(x,z) \le 0 \quad \Longleftrightarrow \quad x \le x_r(z),$$

When $J_U(x,z)$ is below the reservation productivity, the match is destroyed. Destroyed matches return the worker to unemployment, where the worker reoptimises between search and training.

5.5 Job creation

At equilibrium, $V_U = 0$. Free entry implies:

$$J_U(\bar{x}(z),z) = \frac{k_U}{q_U(\theta_U(z))}.$$

Remark: unemployment search value with endogenous training option. When the bargaining outside option is the full unemployment value $U_U(x,z)$, the standard Pissarides simplification no longer yields a closed-form expression for $U_U^{\text{search}}(z)$. However, combining Nash bargaining, free entry, and the search HBJ equation still delivers a useful scalar condition.

Evaluating the search value HBJ equation at the productivity of newly created matches $x = \bar{x}(z)$ and using free entry,

$$J_U(\bar{x}(z),z) = \frac{k_U}{q_U(\theta_U(z))},$$

together with Nash bargaining,

$$E_U(\bar{x}(z),z) = U_U(\bar{x}(z),z) + \frac{\beta_U}{1-\beta_U} J_U(\bar{x}(z),z),$$

yields

$$(r-\mathbf{Q})U_U^{\text{search}}(z) = b_U(z) + \theta_U(z)q_U(\theta_U(z))\Big(U_U(\bar{x}(z),z) - U_U^{\text{search}}(z)\Big) + \frac{\beta_U}{1-\beta_U}k_U\,\theta_U(z). \quad (*)$$

Relative to the benchmark model in which the outside option is $U_U^{\text{search}}(z)$, equation $(*)$ contains an additional term capturing the option value of training at the hiring margin. This term is weakly positive and strictly positive whenever training is optimal for workers with type $x > \bar{x}(z)$.

Computational use (block Gauss–Seidel update). Although equation $(*)$ does not yield a closed-form expression for $U_U^{\text{search}}(z)$, it provides a conditional characterization that can be exploited computationally. For given values of $U_U(\bar{x}(z), z)$ and $\theta_U(z)$, equation $(*)$ is linear in $U_U^{\text{search}}(z)$ and can be solved directly.

In the numerical algorithm, I therefore update $U_U^{\text{search}}(z)$ by solving $(*)$ conditional on the current iterate of $U_U(\bar{x}(z), z)$, and immediately reconstruct $U_U(\cdot, z)$ using the max operator

$$U_U(x, z) = \max\{U_U^{\text{search}}(z), -c(x) + T(x, z)\}.$$

The remaining value functions are then updated using this newly computed $U_U(\cdot, z)$.

This sequential updating scheme constitutes a block Gauss–Seidel step within the inner fixed-point iteration. Relative to treating $U_U^{\text{search}}(z)$ as an independent component of the joint HBJ system, this approach substantially improves convergence by exploiting the conditional structure of the search value.

5.6 Unskilled Equilibrium

Fix a training cutoff function $\bar{x} : \mathcal{Z} \to [0, 1]$ and take as given $U_S(x, z)$. An *unskilled equilibrium given* $\bar{x}(\cdot)$ consists of

$$\Big\{\theta_U(z), x_r(z), U_U(x, z), T(x, z), E_U(x, z), J_U(x, z), w_U(x, z)\Big\}_{z \in \mathcal{Z}}$$

such that for each $z \in \mathcal{Z}$:

(i) Unemployment, training, and employment values satisfy the HBJ equations above.

(ii) Wages satisfy Nash bargaining with outside option $U_U(x, z)$.

(iii) $x_r(z)$ satisfies $J_U(x_r(z), z) = 0$.

(iv) Free entry holds at $x = \bar{x}(z)$.

5.7 Numerical solution: unskilled block with endogenous training option

We now describe the numerical procedure used to solve the unskilled labor market for a given training cutoff function $\bar{x}(\cdot)$ and skilled-unemployment value $U_S(x, z)$.

The key complication relative to the benchmark Pissarides model is that the bargaining outside option is the full unemployment value $U_U(x, z)$, which itself depends on the endogenous training decision. As a result, the closed-form expression for $U_U^{\text{search}}(z)$ is no longer available, and the value functions must be solved jointly.

Step 1: Training value. For each worker type x and aggregate state z, solve the linear system

$$(r - \mathbf{Q})T(x, z) = b_T(z) + \phi\big(U_S(x, z) - T(x, z)\big).$$

This step is independent of unskilled-market tightness and job destruction and can be solved once per evaluation of the unskilled block, conditional on $U_S(x, z)$.

Step 2: Initialization. For each $z \in \mathcal{Z}$, initialize guesses for market tightness $\theta_U(z)$ and reservation productivity $x_r(z)$. Initialize the search value $U_U^{\text{search}}(z)$ and construct the unemployment value

$$U_U(x,z) = \max\left\{U_U^{\text{search}}(z), -c(x) + T(x,z)\right\}.$$

Step 3: Inner loop — value functions. Given $(\theta_U(z), x_r(z))$:

(a) Update $U_U^{\text{search}}(z)$ using the scalar fixed point

$$(r-\mathbf{Q})U_U^{\text{search}}(z) = b_U(z)+\theta_U(z)q_U(\theta_U(z))\Big(U_U(\bar{x}(z),z)-U_U^{\text{search}}(z)\Big)+\frac{\beta_U}{1-\beta_U}k_U\,\theta_U(z),$$

(b) Given $U_U(x,z)$, solve the coupled HBJ equations for $\{E_U(x,z), J_U(x,z)\}_{x\in[x_r(z),\bar{x}(z)]}$ using Nash bargaining to eliminate wages:

$$\beta_U J_U(x,z) = (1-\beta_U)\big(E_U(x,z) - U_U(x,z)\big).$$

(c) Update the unemployment value pointwise via

$$U_U(x,z) = \max\left\{U_U^{\text{search}}(z), -c(x) + T(x,z)\right\}.$$

Steps (a)–(c) are iterated until convergence of the value functions for fixed $(\theta_U(z), x_r(z))$.

Step 4: Job destruction. Update the reservation productivity $x_r(z)$ using the boundary condition

$$J_U(x_r(z), z) = 0.$$

On a discrete grid, this is implemented by selecting the lowest productivity level at which $J_U(x,z) \geq 0$.

Step 5: Job creation. Update market tightness using free entry at the top of the productivity support:

$$J_U(\bar{x}(z), z) = \frac{k_U}{q_U(\theta_U(z))}.$$

Step 6: Outer loop and convergence. Steps 3–5 are iterated until convergence of

$$\big(\theta_U(z), x_r(z), U_U^{\text{search}}(z)\big) \quad \text{for all } z \in \mathcal{Z}.$$

Output. Given convergence, recover the full set of equilibrium objects

$$\left\{U_U(x,z), T(x,z), E_U(x,z), J_U(x,z), \theta_U(z), x_r(z)\right\}_{z\in\mathcal{Z}},$$

and the implied training policy: train if and only if

$$-c(x) + T(x,z) \geq U_U^{\text{search}}(z).$$

6 Skilled Labor Market (LMR 2016)

This section follows Lise et al. (2016) as closely as possible, while adapting their notation to our environment with aggregate state $z \in \mathcal{Z}$ and match output $\pi_S(x, y, z) = P_S(z)f(x, y)$.

6.1 Primitives and market objects

Types. Workers have innate type $x \in [0, 1]$ and skilled jobs/firms have type $y \in [0, 1]$.

Market access and support. The education cutoff $\bar{x}(z)$ partitions workers into markets. Only types $x \geq \bar{x}(z)$ participate in the skilled labor market. Accordingly, skilled worker objects satisfy

$$u_S(x, z) = e_S(x, z) = 0 \quad \text{for } x < \bar{x}(z),$$

and all integrals over worker type in the skilled block are effectively taken over $[\bar{x}(z), 1]$.

Separation and shocks. A filled skilled match is destroyed exogenously at Poisson rate $\xi_S > 0$. In addition, match quality is hit by idiosyncratic "job-type" shocks at Poisson rate $\delta_S > 0$. Upon a shock, the job type is redrawn as

$$y' \sim \Gamma(\cdot) \quad \text{with density } \gamma(\cdot),$$

and the match continues only if the post-shock surplus is nonnegative; otherwise the worker becomes skilled unemployed and the firm posts a vacancy immediately (as in Lise et al. (2016)). When a match ends, the same job switches from filled to vacant; we are not creating an additional job. Net job creation is governed separately by the posting/free-entry condition.

Search. Unemployed search intensity is normalized to 1. Employed workers search on the job with relative intensity $s \in [0, 1]$.

Flow payoff out of work. Skilled unemployment delivers flow payoff $b_S(z)$.

Vacancy posting cost. Posting and maintaining a skilled vacancy of type y costs $k_S(y) \geq 0$ per unit time.

Vacancy and unemployment measures. Let $u_S(x, z)$ denote the density of skilled unemployed workers by type x in state z. Let $v_S(y, z)$ denote the density of skilled vacancies by job type y in state z. Define total masses

$$\mathcal{U}_S(z) = \int_{\bar{x}(z)}^{1} u_S(x, z)\ \mathrm{d}x., \qquad \mathcal{V}_S(z) = \int_0^1 v_S(y, z)\ \mathrm{d}y.$$

Match distribution. Let $h(x, y, z)$ denote the density of ongoing skilled matches between a worker of type x and a job of type y in aggregate state z.

Consistency (marginals). Match density pins down employment marginals:

$$e_S(x, z) \equiv \int_0^1 h(x, y, z)\ \mathrm{d}y, \qquad n_S(y, z) \equiv \int_0^1 h(x, y, z)\ \mathrm{d}x,$$

where $e_S(x, z)$ is the density of employed workers of type x, and $n_S(y, z)$ is the density of filled jobs of type y.

Effective searchers. Let total employed mass be

$$\mathcal{E}_S(z) = \int_{\bar{x}(z)}^{1} \int_0^1 h(x, y, z)\ \mathrm{d}y\ \mathrm{d}x.$$

and define total effective searchers as

$$\widetilde{\mathcal{U}}_S(z) \equiv \mathcal{U}_S(z) + s\,\mathcal{E}_S(z).$$

6.2 Matching and meeting rates

Matching function. Matches are formed according to a constant-returns matching technology

$$M_S\big(\widetilde{\mathcal{U}}_S(z), \mathcal{V}_S(z)\big) = \mu_S\,\widetilde{\mathcal{U}}_S(z)^{\eta_S}\,\mathcal{V}_S(z)^{1-\eta_S}, \qquad \mu_S > 0,\ \eta_S \in (0,1).$$

LMR matching parameter. Following Lise et al. (2016), define

$$\kappa(z) \equiv \frac{M_S\big(\widetilde{\mathcal{U}}_S(z), \mathcal{V}_S(z)\big)}{\widetilde{\mathcal{U}}_S(z)\,\mathcal{V}_S(z)}.$$

This parameter summarizes tightness in a single object that directly scales contact rates.

Meeting rates by type. A (skilled) unemployed worker of type x meets a vacancy of type y at rate

$$\kappa(z)\,v_S(y, z).$$

A vacancy of any type meets an unemployed worker of type x at rate

$$\kappa(z)\,u_S(x, z).$$

An employed worker (of any type) receives contacts at rate

$$s\,\kappa(z)\,v_S(y, z) \quad \text{from vacancies of type } y.$$

Flow of contacts by pair. Given $\kappa(z)$, the flow rate of contacts between unemployed workers of type x and vacancies of type y is

$$\kappa(z)\,u_S(x, z)\,v_S(y, z).$$

A new match is created if and only if $S(x, y, z) \geq 0$, so the inflow into the match distribution is

$$\kappa(z)\,u_S(x, z)\,v_S(y, z)\,\mathbb{1}\{S(x, y, z) \geq 0\}.$$

6.3 Value functions

To keep notation parallel to the unskilled block, we use the same labels: unemployment U_S, employment E_S, vacancy V_S, and filled job J_S.

Unemployed worker. Let $U_S(x, z)$ be the value of skilled unemployment for a worker of type x in aggregate state z.

Vacancy. Let $V_S(y, z)$ be the value of a vacant skilled job of type y in aggregate state z.

Employed worker under contract. Let $E_S(w, x, y, z)$ be the value to a worker of being employed in a feasible match (x, y) at current wage contract w (continuous time; w is a flow wage).

Filled job under contract. Let $J_S(w, x, y, z)$ be the value to the firm of an occupied job (x, y) paying wage w.

Bargaining set and renegotiation triggers. Define the *current worker rent* (relative to unemployment) as

$$R_S(w, x, y, z) \equiv E_S(w, x, y, z) - U_S(x, z).$$

For a feasible match, the total surplus $S(x, y, z)$ (defined below) pins down the bargaining set:

$$0 \leq R_S(w, x, y, z) \leq S(x, y, z).$$

If $R(w, x, y, z)$ exits this interval, renegotiation occurs to the *closest* point in the bargaining set, as in Lise et al. (2016). Specifically, define two contract adjustment schedules:

$$\begin{aligned} \psi_0(x, y, z): &\quad E_S(\psi_0, x, y, z) - U_S(x, z) = 0 \quad \text{(worker participation binds)}, \\ \psi_1(x, y, z): &\quad E_S(\psi_1, x, y, z) - U_S(x, z) = S(x, y, z) \quad \text{(firm participation binds)}. \end{aligned}$$

Outside offers and the set of poaching vacancies. Given current contract (w, x, y, z), define the set of vacancies that can induce a wage change (either via job-to-job move or a credible outside option that forces a counteroffer):

$$A(w, x, y, z) \equiv \Big\{\tilde{y} \in [0, 1] : E_S(w, x, y, z) - U_S(x, z) < S(x, \tilde{y}, z)\Big\}.$$

Let its vacancy mass be

$$v_S(A(w, x, y, z), z) \equiv \int_{A(w,x,y,z)} v_S(\tilde{y}, z)\ \mathrm{d}\tilde{y}.$$

Skilled unemployment (option value). An unemployed worker receives $b_S(z)$ and meets vacancies at rate $\kappa(z)v_S(y, z)$. If a contacted vacancy is feasible ($S(x, y, z) \geq 0$), the worker obtains a share $\beta_S \in (0, 1)$ of the match surplus upon hiring from unemployment. Thus,

$$(r - \mathbf{Q})U_S(x, z) = b_S(z) + \kappa(z) \int_0^1 \beta_S \left(S(x, y, z)\right)^+ v_S(y, z)\ \mathrm{d}y, \qquad (a)^+ \equiv \max\{a, 0\}.$$

Vacancy value. A vacant skilled job of type y in aggregate state z pays the flow cost $k_S(y)$. While vacant, the job type is hit by idiosyncratic shocks at Poisson rate δ_S; upon a shock, the job type is redrawn as $y' \sim \Gamma(\cdot)$ with density $\gamma(\cdot)$. The vacancy meets unemployed workers at rate $\kappa(z)\, u_S(x, z)$ and hires from unemployment, receiving share $(1 - \beta_S)$ of the match surplus. In addition, it meets employed searchers at rate $s\,\kappa(z)$; when a vacancy of type y meets an employed worker currently in match (x, y'), the vacancy can profitably poach only if $S(x, y, z) > S(x, y', z)$, in which case the incremental surplus is $\big(S(x, y, z) - S(x, y', z)\big)^+$.

Therefore,

$$
\begin{aligned}
(r-\mathbf{Q})V_S(y,z) = -k_S(y) &+ \delta_S \int_0^1 \Big(V_S(y',z) - V_S(y,z)\Big)\,\gamma(y')\,\mathrm{d}y' \\
&+ \kappa(z)\int_{\bar{x}(z)}^1 (1-\beta_S)\,\big(S(x,y,z)\big)^+\,u_S(x,z)\,\mathrm{d}x \\
&+ s\,\kappa(z)\,(1-\beta_S)\int_0^1\int_{\bar{x}(z)}^1 \big(S(x,y,z)-S(x,y',z)\big)^+\,h(x,y',z)\,\mathrm{d}x\,\mathrm{d}y'.
\end{aligned}
$$

Interpretation of the poaching term. A crucial implication of the LMR renegotiation protocol is that a vacancy's gain from poaching an employed worker is *not* the full surplus of the new match $S(x,y,z)$. Instead, the worker obtains the entire surplus of the incumbent job plus a share of the incremental surplus between the two jobs.

The bargained wage when switching from job y' to y should satisfy:

$$
\begin{aligned}
E_S(w,x,y,z) - U_S(x,z) &= S(x,y',z) + \beta[S(x,y,z) - S(x,y',z)] \\
&= \beta_S\,S(x,y,z) + (1-\beta_S)\,S(x,y',z),
\end{aligned}
$$

so that the worker retains a share of the old-match surplus in addition to a share of the incremental surplus. This reflects the fact that the worker can credibly threaten to remain in the current match if the poaching firm does not offer sufficiently favorable terms.

Since total surplus in the new match is $S(x,y,z)$, the firm's rent from poaching is therefore

$$
S(x,y,z) - \Big[\beta_S\,S(x,y,z) + (1-\beta_S)\,S(x,y',z)\Big] = (1-\beta_S)\big(S(x,y,z) - S(x,y',z)\big).
$$

This expression is positive only when the new match yields higher surplus than the incumbent match, which explains the truncation by the positive part operator in the vacancy value. Accordingly, the poaching term in the vacancy Bellman equation integrates $(1-\beta_S)\big(S(x,y,z)-S(x,y',z)\big)^+$ over the distribution of existing matches $h(x,y',z)$.

Employed worker (rent formulation; LMR equation (12)). While employed, four events can occur: (i) exogenous separation at rate ξ_S, (ii) a job-type shock at rate δ_S with redraw $y' \sim \Gamma(\cdot)$, (iii) contact with outside vacancies at rate $s\,\kappa(z)$, and (iv) aggregate state transitions governed by $\mathbf{Q}$. Recall $R_S(w,x,y,z) \equiv E_S(w,x,y,z) - U_S(x,z)$. Define the set of outside offers that can alter the worker's position as

$$
A(w,x,y,z) \equiv \{\tilde{y} \in [0,1] : R_S(w,x,y,z) < S(x,\tilde{y},z)\},
$$

and let $v_S(A(w,x,y,z),z)$ denote its vacancy mass i.e. $v_S(A(w,x,y,z),z) = \int_{A(w,x,y,z)} v_S(y,z)dy$. Then the worker's rent satisfies

$$
\begin{aligned}
&\Big[(r-\mathbf{Q}) + \xi_S + \delta_S + s\,\kappa(z)\,v_S(A(w,x,y,z),z)\Big]\,R_S(w,x,y,z) \\
&= \left[w - b_S(z) - \kappa(z)\int_0^1 \beta_S\,\big(S(x,y',z)\big)^+\,v_S(y',z)\,\mathrm{d}y'\right] \\
&\quad + \delta_S\int_0^1 \big[\min\{S(x,y',z),\,R_S(w,x,y',z)\}\big]^+\,\gamma(y')\,\mathrm{d}y' \\
&\quad + s\,\kappa(z)\int_{A(w,x,y,z)} \big[\min\big\{S(x,y,z),S(x,y',z)\big\} + \beta_S\big[S(x,y',z) - S(x,y,z)\big]^+\big]\,v_S(y',z)\,\mathrm{d}y'.
\end{aligned}
$$

Interpretation. This expression mirrors equation (12) in Lise et al. (2016). The first line collects the effective discounting of the current rent due to separation, shocks, and relevant outside offers. The second line is the flow advantage of employment relative to unemployment: the worker earns wage w but forgoes the option value of unemployment search. The third line reflects job-type shocks: when y redraws to y', the existing contract may violate the new bargaining set, in which case the worker's rent is reset to the closest feasible point, represented by truncation at $S(x, y', z)$. The final line captures outside offers. Only vacancies with surplus exceeding the worker's current rent are relevant. Upon contact, Bertrand competition between the incumbent and the poaching firm implies that the worker's continuation rent is pinned down by the relative surpluses of the two matches, subject to feasibility. As in the vacancy problem, this mechanism ensures that rents adjust endogenously in response to outside options without requiring explicit wage schedules.

Joint value of a skilled match. Following Appendix A of Lise et al. (2016), it is convenient to define the *joint value* of a skilled match between a worker of type x and a job of type y in aggregate state z. Denote this object by $\Pi_S(x, y, z)$. It represents the present value of total production generated by the match, before any division of surplus between the worker and the firm. Since wages are pure transfers, $\Pi_S(x, y, z)$ is independent of the current wage contract and depends only on primitives, outside options, and equilibrium continuation rules.

The joint value satisfies the Bellman equation

$$
\begin{aligned}
(r - \mathbf{Q})\,\Pi_S(x, y, z) = {} & P_S(z) f(x, y) \\
& + \xi_S\Big(U_S(x, z) + V_S(y, z) - \Pi_S(x, y, z)\Big) \\
& + s\,\kappa(z) \int_0^1 \Bigg[\max\Big\{\Pi_S(x, y, z), \\
& U_S(x, z) + V_S(y, z) + S(x, y, z) + \beta_S\big(S(x, \tilde{y}, z) - S(x, y, z)\big)\Big\} - \Pi_S(x, y, z)\Bigg] v_S(\tilde{y}, z)\ \mathrm{d}\tilde{y} \\
& + \delta_S \int_0^1 \Big[\max\Big\{\Pi_S(x, y', z),\ U_S(x, z) + V_S(y', z)\Big\} - \Pi_S(x, y, z)\Big]\, \gamma(y')\ \mathrm{d}y'.
\end{aligned}
$$

The first term is current match output. The second term reflects exogenous separation, in which case the joint value collapses to the sum of the worker's unemployment value and the value of a vacant job. The third term captures job-to-job contacts: when an employed worker meets an outside vacancy, Bertrand competition and feasibility constraints determine whether the match continues and at what continuation value. The final term reflects idiosyncratic job-type shocks, which may lead either to continuation of the match or to endogenous separation if joint surplus becomes negative.

6.4 Surplus formulation

Definition. Define the total surplus of a skilled match between a worker of type x and a job of type y in aggregate state z as the joint value of the match net of outside options:

$$S(x, y, z) \;\equiv\; \Pi_S(x, y, z) - U_S(x, z) - V_S(y, z).$$

By construction, $S(x, y, z)$ is independent of the current wage contract. A match is feasible if and only if $S(x, y, z) \geq 0$.

Interpretation. The surplus measures the total gains from trade generated by the match relative to separation, where separation delivers unemployment for the worker and a vacant job for the firm. Since wages are pure transfers, the surplus depends only on primitives, search frictions, and equilibrium continuation rules. As in Lise et al. (2016), all equilibrium objects can be characterized in terms of $S(x, y, z)$.

Surplus fixed point (LMR). Define the skilled match surplus by

$$S(x, y, z) \equiv \Pi_S(x, y, z) - U_S(x, z) - V_S(y, z).$$

Substituting the definitions of $P_S(x, y, z)$, $U_S(x, z)$, and $V_S(y, z)$ into the joint-value Bellman equation yields a closed recursion for the surplus. Following Appendix A of Lise et al. (2016), $S(x, y, z)$ is independent of the current wage contract and satisfies the fixed point

$$\begin{aligned}
\Big[(r - \mathbf{Q}) + \xi_S + \delta_S\Big] S(x, y, z) = P_S(z) f(x, y) - b_S(z) + k_S(y) \\
- \kappa(z) \int_0^1 \beta_S \left(S(x, \tilde{y}, z)\right)^+ v_S(\tilde{y}, z) \, \mathrm{d}\tilde{y} \\
- \kappa(z) \int_{\bar{x}(z)}^1 (1 - \beta_S) \left(S(\tilde{x}, y, z)\right)^+ u_S(\tilde{x}, z) \, \mathrm{d}\tilde{x} \\
- s\,\kappa(z) \int_0^1 \int_{\bar{x}(z)}^1 (1 - \beta_S) \Big(S(\tilde{x}, y, z) - S(\tilde{x}, \tilde{y}, z)\Big)^+ h(\tilde{x}, \tilde{y}, z) \, \mathrm{d}\tilde{x} \, \mathrm{d}\tilde{y} \\
+ s\,\kappa(z) \int_0^1 \beta_S \Big(S(x, \tilde{y}, z) - S(x, y, z)\Big)^+ v_S(\tilde{y}, z) \, \mathrm{d}\tilde{y} \\
+ \delta_S \int_0^1 \left(S(x, y', z)\right)^+ \gamma(y') \, \mathrm{d}y'.
\end{aligned}$$

Discussion. The first term is current net output relative to unemployment. The second term reflects the loss of surplus due to exogenous separation and job-type shocks. The third term captures continuation following productivity shocks: the match survives only if post-shock surplus is nonnegative. The final term reflects job-to-job competition. When an employed worker meets a vacancy offering higher surplus, the incremental gains from trade are realized, while otherwise the match continues unchanged. This formulation corresponds exactly to the surplus fixed point in Lise et al. (2016), extended to allow for aggregate state dynamics through $\mathbf{Q}$.

6.5 Free entry

Free entry condition. Under endogenous vacancy posting, free entry implies $V_S(y, z) = 0$

$$\begin{aligned}
k_S(y) = \delta_S \int_0^1 \Big(V_S(y', z) - V_S(y, z)\Big) \gamma(y') \, \mathrm{d}y' \\
+ \kappa(z) \int_0^1 (1 - \beta_S) \left(S(x, y, z)\right)^+ u_S(x, z) \, \mathrm{d}x \\
+ s\,\kappa(z)\,(1 - \beta_S) \int_0^1 \int_{\bar{x}(z)}^1 \left(S(x, y, z) - S(x, y', z)\right)^+ h(x, y', z) \, \mathrm{d}x \, \mathrm{d}y'.
\end{aligned}$$

Implementation note (LMR algorithm). As in Appendix B of Lise et al. (2016), equilibrium can be computed without iterating on wage contracts or on the value functions $E_S(\cdot)$ and $J_S(\cdot)$. The key observation is that the economy is fully characterized by the surplus function $S(x, y, z)$, market tightness $\kappa(z)$, and the induced distributions $\{u_S(x, z), v_S(y, z), h(x, y, z)\}$.

For a given aggregate state z, the computational procedure is as follows. First, guess a value for market tightness $\kappa(z)$. Second, solve the surplus fixed point for $S(x, y, z)$. This determines the set of feasible matches $\{(x, y) : S(x, y, z) \geq 0\}$ as well as the gains from poaching and continuation. Third, impose free entry using the condition above to determine which job types y are actively posted. Fourth, construct the match distribution $h(x, y, z)$ and unemployment distribution $u_S(x, z)$ from the implied flow balance between job creation, destruction, and job-to-job transitions. Finally, update $\kappa(z)$ so that matching flows are consistent with the implied masses of effective searchers and vacancies.

This procedure is iterated until convergence. At no point is it necessary to compute equilibrium wage schedules or contract-specific value functions.

7 Equilibrium and Solution Algorithm

This section describes a practical nested fixed-point algorithm to compute an equilibrium of the full model with endogenous training (market-access cutoff) and segmented search markets. The algorithm treats the unskilled and skilled blocks as coupled through (i) the skilled-unemployment value $U_S(x, z)$ entering the training problem, and (ii) the cutoff function $\bar{x}(z)$ that partitions the worker population and truncates distributions in both blocks.

7.1 Objects and equilibrium conditions

An equilibrium consists of functions and scalars

$$\Big\{\bar{x}(z),\, \theta_U(z),\, x_r(z),\, U_U^{\text{search}}(z)\Big\}_{z\in\mathcal{Z}}, \qquad \Big\{U_U(x, z), T(x, z), E_U(x, z), J_U(x, z)\Big\}_{x\in[0,\bar{x}(z)]},$$

and skilled-market objects

$$\Big\{\kappa(z),\, S(x, y, z),\, u_S(x, z),\, v_S(y, z),\, h(x, y, z),\, U_S(x, z)\Big\}_{x\in[\bar{x}(z),1],\, y\in[0,1],\, z\in\mathcal{Z}},$$

such that: (i) the unskilled block is an equilibrium given $\bar{x}(\cdot)$ and $U_S(\cdot, \cdot)$, (ii) the skilled block is an equilibrium given $\bar{x}(\cdot)$, and (iii) the cutoff condition holds for each z:

$$U_U^{\text{search}}(z) = -c(\bar{x}(z)) + T(\bar{x}(z), z),$$

with training optimal for $x > \bar{x}(z)$ and unskilled search optimal for $x \leq \bar{x}(z)$.

7.2 Numerical algorithm (nested fixed points)

Let $\varepsilon > 0$ be a convergence tolerance. The procedure iterates over the vector of cutoffs $\bar{x}(\cdot)$ and the two market-tightness objects $\theta_U(\cdot)$ and $\kappa(\cdot)$.

Step 0: Initialization. Choose initial guesses:

$$\bar{x}^{(0)}(z) \in (0, 1), \quad \theta_U^{(0)}(z) > 0, \quad \kappa^{(0)}(z) > 0 \qquad \forall z \in \mathcal{Z}.$$

Initialize skilled values $U_S^{(0)}(x, z)$ on $x \in [\bar{x}^{(0)}(z), 1]$ (e.g. constant or from a simplified model). Set outer iteration counter $m = 0$.

Step 1: Skilled block given $\bar{x}^{(m)}(\cdot)$. For each aggregate state $z \in \mathcal{Z}$ (or jointly across z if you solve with the operator $(r - \mathbf{Q})$), compute a skilled-market equilibrium conditional on $\bar{x}^{(m)}(z)$:

(S1) **Given** $\kappa^{(m)}(z)$ and current distributions, solve the surplus fixed point for $S^{(m)}(x, y, z)$ on $x \in [\bar{x}^{(m)}(z), 1]$, $y \in [0, 1]$ using the LMR recursion.

(S2) **Free entry / active vacancies.** Impose $V_S(y, z) = 0$ and the free-entry condition to identify the set of job types with positive posting and to construct (or update) the vacancy density $v_S^{(m)}(y, z)$.

(S3) **Stationary flow balance.** Given feasibility $\mathbb{1}\{S^{(m)}(x, y, z) \geq 0\}$ and the LMR poaching logic, construct the match density $h^{(m)}(x, y, z)$ and unemployment density $u_S^{(m)}(x, z)$ from flow balance (job creation, endogenous/exogenous destruction, and job-to-job transitions).

(S4) **Update market tightness.** Compute implied totals

$$\mathcal{U}_S^{(m)}(z) = \int_{\bar{x}^{(m)}(z)}^1 u_S^{(m)}(x, z)\, \mathrm{d}x, \quad \mathcal{E}_S^{(m)}(z) = \int_{\bar{x}^{(m)}(z)}^1 \int_0^1 h^{(m)}(x, y, z)\, \mathrm{d}y\, \mathrm{d}x,$$

$$\mathcal{V}_S^{(m)}(z) = \int_0^1 v_S^{(m)}(y, z)\, \mathrm{d}y,$$

form $\widetilde{\mathcal{U}}_S^{(m)}(z) = \mathcal{U}_S^{(m)}(z) + s\, \mathcal{E}_S^{(m)}(z)$, and update

$$\kappa^{(m+1)}(z) \;\leftarrow\; \frac{M_S(\widetilde{\mathcal{U}}_S^{(m)}(z), \mathcal{V}_S^{(m)}(z))}{\widetilde{\mathcal{U}}_S^{(m)}(z)\, \mathcal{V}_S^{(m)}(z)},$$

(with damping if needed).

(S5) **Update U_S.** Given $S^{(m)}$ and $v_S^{(m)}$, update skilled unemployment values via

$$(r - \mathbf{Q})U_S^{(m+1)}(x, z) = b_S(z) + \kappa^{(m+1)}(z) \int_0^1 \beta_S\, (S^{(m)}(x, y, z))^+\, v_S^{(m)}(y, z)\, \mathrm{d}y.$$

Iterate (S1)–(S5) until convergence of $\{S, \kappa, u_S, v_S, h, U_S\}$ for the given cutoff.

Denote the converged skilled-unemployment value by $U_S^\star(x, z; \bar{x}^{(m)})$.

Step 2: Unskilled block given $\bar{x}^{(m)}(\cdot)$ and $U_S^\star$. For each aggregate state $z \in \mathcal{Z}$ (or jointly across z), solve the unskilled equilibrium conditional on $\bar{x}^{(m)}(z)$ and $U_S^\star(\cdot, z)$ using the inner/outer scheme already stated:

(U1) Compute training values $T^\star(x, z)$ from

$$(r - \mathbf{Q})T(x, z) = b_T(z) + \phi\big(U_S^\star(x, z) - T(x, z)\big).$$

(U2) Initialize $(\theta_U(z), x_r(z), U_U^{\text{search}}(z))$ and set

$$U_U(x, z) = \max\{U_U^{\text{search}}(z), -c(x) + T^\star(x, z)\}.$$

(U3) Iterate the unskilled inner fixed point for value functions (Gauss–Seidel): update $U_U^{\text{search}}(z)$ from $(*)$, then update (E_U, J_U) using Nash bargaining to eliminate wages, then update $U_U(\cdot, z)$ via the max operator.

(U4) Update job destruction using $J_U(x_r(z), z) = 0$ (grid implementation: first index where $J_U \geq 0$).

(U5) Update job creation using free entry at $\bar{x}^{(m)}(z)$:

$$J_U(\bar{x}^{(m)}(z), z) = \frac{k_U}{q_U(\theta_U(z))}.$$

Iterate (U3)–(U5) until convergence of $\{\theta_U, x_r, U_U^{\text{search}}, U_U, E_U, J_U\}$.

Denote the converged search value by $U_{U,\star}^{\text{search}}(z; \bar{x}^{(m)})$.

Step 3: Cutoff update. For each $z \in \mathcal{Z}$, update the market-access cutoff by solving the indifference condition

$$g_z(x) \equiv -c(x) + T^\star(x, z) - U_{U,\star}^{\text{search}}(z; \bar{x}^{(m)}) = 0 \quad \text{for } x \in [0, 1].$$

Let the solution be $\bar{x}^{\text{new}}(z)$. Apply damping:

$$\bar{x}^{(m+1)}(z) \leftarrow (1 - \omega)\bar{x}^{(m)}(z) + \omega\, \bar{x}^{\text{new}}(z), \qquad \omega \in (0, 1].$$

Step 4: Convergence test. Compute

$$\Delta^{(m)} \equiv \max_{z \in \mathcal{Z}} \left|\bar{x}^{(m+1)}(z) - \bar{x}^{(m)}(z)\right|.$$

If $\Delta^{(m)} < \varepsilon$, stop. Otherwise set $m \leftarrow m + 1$ and return to Step 1.

Outputs. Upon convergence, report:

$$\Big\{\bar{x}(z), \theta_U(z), x_r(z), U_U(\cdot, z), T(\cdot, z), E_U(\cdot, z), J_U(\cdot, z)\Big\}_{z \in \mathcal{Z}},$$

and

$$\Big\{\kappa(z), S(\cdot, \cdot, z), u_S(\cdot, z), v_S(\cdot, z), h(\cdot, \cdot, z), U_S(\cdot, z)\Big\}_{z \in \mathcal{Z}},$$

together with the implied training policy

$$\text{train in } (x, z) \iff -c(x) + T(x, z) \geq U_U^{\text{search}}(z).$$